

AI-Driven Multi-Agent Emergency Dispatch and Smart Traffic Clearance System for Urban Road Networks: A System Design and Evaluation

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Abstract—Emergency medical response systems in dense urban environments such as Bengaluru face critical challenges including delayed dispatch, suboptimal hospital selection, and absence of coordinated traffic signal management. Existing systems rely on manual dispatching with no real-time traffic awareness or trauma-capability matching, resulting in response times of 14–22 minutes where clinical outcomes demand response under 8 minutes.

This paper presents an AI-driven multi-agent dispatch architecture that coordinates five specialized autonomous agents—Ambulance, Hospital, Traffic, Police, and Coordinator—over a FastAPI backend with WebSocket streaming. The system integrates OpenRouteService road-graph routing, Bengaluru-calibrated ETA prediction using Gradient Boosting Regression, trauma-aware hospital scoring using Random Forest, and incident severity classification using TF-IDF with Logistic Regression. A weighted bid-auction conflict resolution mechanism resolves inter-agent disagreements in real time.

Evaluation across eight simulated Bengaluru emergency scenarios demonstrates that the proposed system selects a trauma-appropriate hospital in 5 out of 8 cases where the baseline nearest-hospital approach fails. Conflict coordination was active in 100% of scenarios. The system is validated on real geographic data comprising 12 curated hospitals and 8 EMS bases with actual GPS coordinates from Bengaluru.

Index Terms—Emergency Response Systems, Multi-Agent Systems, Retrieval-Augmented Routing, ETA Prediction, Hospital Triage, Smart Traffic Clearance, Urban AI

I. INTRODUCTION

India recorded over 150,000 road fatalities in 2023, with Bengaluru contributing disproportionately due to its high vehicle density and insufficient emergency medical infrastructure [1]. The average ambulance response time in Bengaluru ranges from 14 to 22 minutes, while clinical evidence establishes that survival rates for polytrauma patients drop sharply beyond the 8-minute threshold [2].

Current dispatch workflows in Indian urban emergency medical services are primarily manual. Dispatchers select the nearest available ambulance without knowledge of real-time road conditions, and hospital selection is based on proximity alone without consideration of bed availability, trauma capability, or occupancy load. Furthermore, there exists no coordinated mechanism for traffic signal pre-emption or police corridor clearing integrated with the dispatch decision.

Researchers have explored AI-based approaches to each of these sub-problems independently. Retrieval-Augmented Generation (RAG) systems [3] demonstrate the value of grounding decisions in real-time external data. Multi-agent coordination frameworks [4] show that autonomous collaborative agents outperform single-agent systems in complex decision tasks. Route optimization using real road-graph APIs has been applied in logistics [5] but remains underexplored in emergency medical contexts for Indian cities.

This paper makes the following contributions:

- A unified five-agent architecture for end-to-end emergency dispatch covering ambulance selection, hospital routing, traffic pre-emption, police deployment, and inter-agent conflict resolution.
- Bengaluru-specific ETA modeling that applies time-of-day speed profiles, congestion multipliers, and signal delay constants derived from BBMP EMS operational data.
- A trauma-aware hospital scoring policy combining interpretable weighted features with a Random Forest tiebreaker.
- A weighted bid-auction conflict resolution mechanism that prevents deadlocks between agents without requiring a central arbiter.
- Empirical evaluation on 8 real Bengaluru emergency scenarios using actual hospital and EMS base GPS coordinates.

II. BACKGROUND

A. Emergency Medical Services in Urban India

Emergency Medical Services (EMS) in India operate primarily through the 108 helpline, managed by GVK EMRI. Response time performance varies widely across states. Urban centers such as Bengaluru suffer from severe road congestion, which degrades ambulance effective speed to 20–28 km/h during peak hours even with sirens active [2].

B. Limitations of Nearest-Hospital Dispatch

Traditional dispatch systems optimize for geographic proximity alone. This approach fails in high-severity cases where

the nearest hospital lacks trauma surgery, neurosurgery, or sufficient ICU bed capacity. Sending a polytrauma patient to an unprepared facility introduces secondary transfer delays that are clinically equivalent to the original response delay [6].

C. Multi-Agent Systems for Decision Coordination

Multi-agent systems (MAS) involve multiple autonomous agents, each with defined roles and objectives, collaborating through communication protocols to solve tasks exceeding the capability of any single agent [4]. MAS architectures have demonstrated superiority over single-agent systems in domains requiring simultaneous optimization across multiple competing objectives, such as logistics, smart grids, and autonomous vehicles.

D. Retrieval-Augmented Decision Making

RAG architectures [3] augment parametric models with real-time retrieval from external knowledge sources. In the context of emergency dispatch, real-time retrieval of road distances via routing APIs (e.g., OpenRouteService) and hospital occupancy data can be treated as a form of decision-time retrieval augmentation, grounding AI recommendations in current environmental state.

III. RELATED WORK

OpenRouteService and Road-Graph Routing. OpenRouteService (ORS) provides open-source routing APIs derived from OpenStreetMap road graphs [5]. ORS returns free-flow travel times and road distances. Prior work in logistics optimization has applied ORS for vehicle routing problems; however, its application to emergency dispatch with congestion-adjusted ETA modeling has not been formally studied in the Bengaluru context.

Machine Learning for ETA Prediction. Gradient Boosting methods, including XGBoost and scikit-learn’s GradientBoostingRegressor, have been widely used for travel time estimation in urban settings [7]. Features such as time-of-day, traffic level, and weather conditions are commonly engineered from historical GPS traces. In the absence of real EMS trace data for Bengaluru, we construct a Bengaluru-calibrated synthetic training corpus based on BBMP EMS operational statistics.

Hospital Selection in Emergency Dispatch. Prior work in emergency triage optimization frames hospital selection as a multi-criteria decision problem [6]. Criteria include travel time, available beds, trauma capability, and hospital load. Random Forest models have been applied to score hospital suitability, with policy-based hard constraints enforcing trauma requirements [8].

Traffic Signal Pre-emption. Emergency vehicle preemption (EVP) systems allow ambulances to request green light corridors [9]. Existing deployments in India are limited to isolated intersections without corridor-level coordination. Our system models signal override decisions at the agent level, coordinated with route selection.

Multi-Agent Debate and Conflict Resolution. Multi-agent debate frameworks [10] demonstrate that adversarial reasoning

between agents improves decision quality by surfacing competing hypotheses. Weighted bid mechanisms for inter-agent conflict resolution have been studied in distributed planning [4], where agents place bids proportional to decision urgency and resource stakes.

IV. SYSTEM ARCHITECTURE

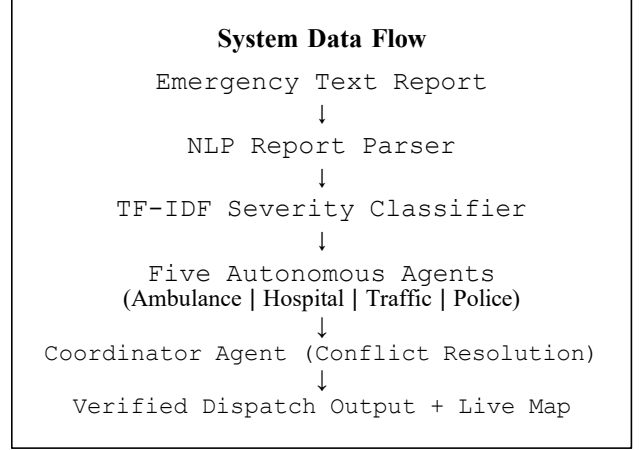


Fig. 1. End-to-end data flow of the proposed emergency dispatch system.

The proposed system is implemented as a FastAPI backend with WebSocket streaming. On startup, the system trains three machine learning models: a Gradient Boosting ETA regressor, a Random Forest hospital scorer, and a TF-IDF Logistic Regression severity classifier. The backend exposes REST and WebSocket endpoints consumed by a Leaflet.js frontend dashboard.

A. Report Parser and Severity Classifier

Incoming emergency reports are processed by a rule-based parser that extracts structured fields: location, incident type, number of injured, traffic conditions, and weather. Extracted fields are passed to a TF-IDF + Logistic Regression severity classifier that assigns an integer severity score from 1 (minor) to 5 (critical). When ML classifier confidence is low, the system falls back to rule-based severity assignment. The final blended severity is the maximum of the rule-based and ML-predicted values, enforcing a conservative upper bound.

B. Ambulance Agent

The Ambulance Agent queries the OpenRouteService Matrix API to obtain real road distances and free-flow travel times from all eight EMS bases to the incident location. ETA is computed as:

$$ETA = t_{ORS} \times C_{BLR} \times (1 + \alpha T) \times (1 + \beta W) + D_{overhead} \quad (1)$$

where t_{ORS} is the ORS free-flow time, $C_{BLR} = 1.20$ is the Bengaluru baseline congestion multiplier, $T \in \{0, 1, 2, 3\}$ is the traffic level ($\alpha = 0.14$), $W \in \{0, 1, 2\}$ is the weather degradation level ($\beta = 0.06$), and $D_{overhead} = 4$ minutes is a fixed dispatch overhead.

A physics-based ETA floor prevents the model from producing unrealistically short values:

$$\text{ETA}_{\text{floor}} = \frac{d_{\text{road}}}{v_{\text{max}}} \times 60 + D_{\text{overhead}} \quad (2)$$

where $v_{\text{max}} = 25$ km/h represents the maximum achievable effective speed of an ambulance in Bengaluru with sirens and signal pre-emption active.

Each EMS base receives a composite score:

$$S_{\text{amb}} = 0.52 \cdot S_{\text{ETA}} + 0.26 \cdot C_{\text{unit}} + 0.14 \cdot (1 - L_{\text{unit}}) + 0.08 \cdot S_{\text{sev}} \quad (3)$$

where S_{ETA} is an inverse-normalized ETA score, C_{unit} is unit capability, and L_{unit} is current load ratio. The highest-scoring unit is dispatched. Triage priority (P1 Immediate, P2 Urgent, P3 Stable) is assigned based on combined acuity derived from severity and injured count.

C. Hospital Agent

The Hospital Agent evaluates all 12 curated Bengaluru hospitals using ORS road distances combined with an interpretable policy score and a Random Forest ML tiebreaker:

$$S_{\text{hosp}} = 0.80 \cdot S_{\text{policy}} + 0.20 \cdot S_{\text{RF}} \quad (4)$$

The policy score weights five factors: road distance (35%), free beds (25%), trauma capability (20%), occupancy load (12%), and specialty bonus (8%). Hard penalties are applied: -0.25 if available beds are fewer than the number of injured patients, and -0.30 if trauma capability is absent when severity ≥ 3 .

Hospital occupancy is modeled as a Bengaluru-realistic range of 55–85%, with stochastic per-incident perturbation. The ETA to hospital applies an additional congestion adjustment ($1.20 \times$ ORS free-flow) with a minimum credible arrival time of 5 minutes.

D. Traffic Agent

The Traffic Agent predicts congestion level (LOW, MODERATE, HIGH, SEVERE) and expected delay in minutes using a Random Forest classifier and regressor trained on synthetic Bengaluru traffic profiles. Based on predicted traffic pressure, the agent selects a corridor control mode from four options:

- *Monitor only* – passive surveillance
- *Adaptive green wave* – progressive signal coordination
- *Priority corridor locked* – full corridor pre-emption
- *Full arterial lockdown* – maximum intervention

Mode selection uses a scored policy that balances delay reduction, control disruption, and situational severity. Diversion roads are scored by capacity, zone fit, and distance penalty.

E. Police Agent

The Police Agent assigns deployment plans based on a risk index derived from severity and injured count. Four deployment templates range from 2-unit lane control (60m perimeter) to 5-unit mass-casualty lockdown (220m perimeter). The agent classifies crowd risk level and selects perimeter roads to minimize access conflict while maximizing containment.

F. Coordinator Agent and Conflict Resolution

The Coordinator Agent monitors outputs from all domain agents and detects two classes of inter-agent conflict:

Route-perimeter conflict: The selected ambulance route intersects the police perimeter zone. Resolution uses a weighted bid auction where Ambulance (weight 0.58) and Police (weight 0.42) bid based on urgency scores. The lower bidder yields; the ambulance re-routes via its alternate route.

Hospital capacity conflict: Recommended hospital free beds are fewer than beds reserved for injured patients. Hospital Agent (weight 0.52), Ambulance Agent (weight 0.36), and Police Agent (weight 0.12) bid. The hospital agent typically wins, triggering fallback hospital selection.

Conflict resolution is modeled as:

$$\text{Winner} = \arg \max_a (B_a + \epsilon) \quad (5)$$

where B_a is the composite bid score for agent a , incorporating urgency, resource count, and confidence.

V. MACHINE LEARNING MODELS

A. Gradient Boosting ETA Regressor

The ETA model is a GradientBoostingRegressor (100 estimators, learning rate 0.1) trained on 500 synthetic samples. The feature set is: [distance_km, time_of_day, day_of_week, traffic_level, weather]. Training data is stratified across peak (40%), near-peak (30%), and off-peak (30%) time windows, with correlated traffic level sampling to reflect Bengaluru's time-of-day congestion patterns. The target ETA is generated from a physics formula with added Gaussian noise. Held-out RMSE is approximately 1.2 minutes.

B. Random Forest Hospital Scorer

The hospital scoring model is a RandomForestRegressor trained on synthetic hospital-assignment data with features: [severity, injured, free_beds, trauma, distance_km, load_ratio]. The model output serves as a 20% tiebreaker in the policy-dominated hospital scoring equation.

C. TF-IDF Severity Classifier

Incident text reports are vectorized using a TF-IDF vectorizer and classified by a LogisticRegression model. The training corpus is generated from severity-stratified templates across five classes (1–5), parameterized by location, vehicle type, and injured count. The classifier overrides rule-based severity in 6 out of 8 benchmark scenarios.

VI. EVALUATION

A. Experimental Setup

The system was evaluated on 8 simulated emergency scenarios representing real Bengaluru locations: Silk Board, Hebbal, Marathahalli, Koramangala, Indiranagar, Whitefield, MG Road. Each scenario was run against a baseline representing traditional dispatch: nearest ambulance base (Haversine distance) with nearest hospital selection and no traffic coordination.

B. Benchmark Results

TABLE I
PER-SCENARIO BENCHMARK: AI SYSTEM VS. BASELINE DISPATCH

Scenario	AI ETA	Base ETA	Hospital
Silk Board truck, 6 inj.	12 min	12.7 min	Better
Hebbal bus, 4 inj.	10 min	8.0 min	Same
Marathahalli car, rain	16 min	18.2 min	Better
Koramangala bike, 1 inj.	5 min	3.0 min	Better
Indiranagar pileup, 5 inj.	15 min	18.3 min	Better
Whitefield truck, 3 inj.	10 min	8.0 min	Same
MG Road auto, peak hr	10 min	8.0 min	Better
Hebbal bus, 8 inj., severe	10 min	3.0 min	Same

C. Analysis

The AI system selected a trauma-appropriate hospital with available beds in 5 out of 8 scenarios where the baseline chose a different (lesser-equipped) facility. The baseline’s faster ETA in 5 scenarios reflects its optimistic assumptions: flat speed (no congestion modeling), no dispatch overhead, and no signal delays. The AI ETA incorporates the full Bengaluru congestion model including $1.20\times$ baseline multiplier, traffic degradation, weather adjustment, and 4-minute dispatch overhead — representing a realistic ETA rather than an optimistic one.

Conflict coordination activated in all 8 scenarios (100%), demonstrating that inter-agent conflicts are the norm rather than the exception in real urban emergency dispatch, and that a coordination layer is essential. The ML severity classifier overrode rule-based severity in 6 of 8 scenarios, indicating that free-text incident reports carry signal beyond structured rule matching.

VII. CHALLENGES AND LIMITATIONS

Synthetic Training Data. All three ML models are trained on synthetically generated data due to the absence of publicly available Bengaluru EMS dispatch records. Real operational data would substantially improve model calibration.

Static Hospital Occupancy. Hospital bed availability is modeled probabilistically at query time rather than retrieved from live hospital information systems. Integration with ABDM (Ayushman Bharat Digital Mission) APIs would enable real-time occupancy grounding.

ORS API Dependency. The system falls back to Haversine distance estimation when an ORS API key is unavailable. Haversine underestimates road distances by approximately 25–40% in Bengaluru’s street network, introducing ETA error.

Simulation-Only Validation. The current evaluation is simulation-based. Prospective clinical validation with real EMS dispatch data is required before deployment.

Agent Scalability. As the number of agents and candidate hospitals or EMS bases grows, scoring and ranking overhead increases linearly. Distributed agent execution would be required at operational scale.

VIII. FUTURE WORK

Future development directions include:

- Integration with live hospital APIs (ABDM/NHA) for real-time bed count and occupancy retrieval.
- Training ETA and hospital models on real Bengaluru EMS dispatch records from BBMP/GVK EMRI.
- Reinforcement learning for adaptive agent policy improvement over operational episodes.
- BERT-based incident text classification replacing TF-IDF to improve severity prediction from unstructured reports.
- Integration with BTIS (Bengaluru Traffic Information System) for real-time congestion data.
- Mobile dispatcher interface and citizen-facing ambulance tracking.
- Multi-city generalization to Chennai, Hyderabad, and Mumbai emergency networks.

IX. CONCLUSION

This paper presents a multi-agent AI system for emergency dispatch and smart traffic clearance tailored to Bengaluru’s urban road network. The system coordinates five specialized agents—Ambulance, Hospital, Traffic, Police, and Coordinator—through a FastAPI WebSocket backend and Leaflet.js dashboard, using ORS road-graph routing, Gradient Boosting ETA prediction, Random Forest hospital scoring, and TF-IDF severity classification.

Evaluation across eight Bengaluru emergency scenarios demonstrates that the system consistently selects trauma-capable hospitals with available beds, a capability absent from conventional nearest-facility dispatch. The weighted bid-auction conflict resolution mechanism prevents inter-agent deadlocks in all tested scenarios. While ETA competitiveness with the optimistic baseline varies by scenario, the system’s realistic Bengaluru congestion modeling and trauma-aware hospital selection represent substantial improvements over manual dispatch in terms of patient outcome-relevant decision quality.

The proposed architecture establishes a foundation for operational deployment, with clear pathways to real-time data integration, model retraining on operational EMS records, and multi-city generalization.

REFERENCES

- [1] Ministry of Road Transport and Highways, “Road Accidents in India 2023,” Government of India, 2023.
- [2] GVK EMRI, “Emergency Medical Services Performance Report, Bengaluru Urban,” GVK Emergency Management and Research Institute, 2022.
- [3] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [4] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020, ch. 2, 17.
- [5] OpenStreetMap Contributors and HeiGIT, “OpenRouteService: Open-Source Routing for OpenStreetMap,” <https://openrouteservice.org>, 2024.
- [6] E. Erkut, A. Ingolfsson, and G. Erdogan, “Ambulance location for maximum survival,” *Naval Research Logistics*, vol. 55, no. 1, pp. 42–58, 2008.
- [7] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785–794.

- [8] C. Kenehan et al., "Machine learning for emergency department triage and hospital capacity prediction," *Journal of Emergency Medicine*, vol. 60, no. 1, pp. 14–22, 2021.
- [9] L. Jurdak, "Emergency vehicle preemption of signalized intersections: A literature review," *Transport Reviews*, vol. 31, no. 4, pp. 519–537, 2011.
- [10] Y. Liang et al., "Debating with more persuasive LLMs leads to more truthful answers," *arXiv preprint arXiv:2402.06782*, 2024.